Name	Owen Hutflesz	**Version**	A B C D E
ID	Student ID	**Other**	
Section		**Marking Instructions**	
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CSE 16-1 Spring 2026 Midterm # 1

April 28th, 2026

Topic	Points	Section	Points
Part A: Nested Quantifiers	6	Part B: Set & Logic	16
Part C: Counting	20	Part D:Truth Table	10
		Total	**52**

Put the version of the exam on top of the bubble sheet. For questions with multiple choice answers, please fill in the bubble on the bubble sheet with a dark colored pen.

Part A: Nested Quantifiers

· ANSWER

Work: 'for any A ... for any B' = 2 universal quantifiers: ·A·U, ·B·U
|P(A∩B)| = 2^|A∪B| and |A·B| = |A|+|B|-|A·B| by incl-excl · (D) ·
(A) wrong: uses · not ·. (B) wrong: · not ·. (C) wrong: no P().

[6 points]

1. For this English statement, find the mathematical statement that corresponds to it: for any set A in the universe U and any set B in the universe U, the cardinality of the power set of the union of A and B is 2 to the power of the cardinality of A plus the cardinality of B minus the cardinality of the intersection of A and B.

   (A) $\forall A \subseteq U, \forall B \subseteq U, |\mathcal{P}(A \cap B)| = 2^{|A|+|B|-|A \cup B|}$

   (B) $\exists A \subseteq U, \exists B \subseteq U, |\mathcal{P}(A \cup B)| = 2^{|A|+|B|-|A \cap B|}$

   (C) $\forall A \subseteq U, \forall B \subseteq U, |A \cup B| = 2^{|A|+|B|-|A \cap B|}$

   (D) $\forall A \subseteq U, \forall B \subseteq U, |\mathcal{P}(A \cup B)| = 2^{|A|+|B|-|A \cap B|}$

   (E) None of the above

· ANSWER

2. For natural numbers n and k, we are given the set $A(n,k)$ of non-decreasing sequences of n natural numbers $a_1 \leq a_2 \leq \ldots \leq a_n$ such that $a_n = k$. What is the negation of the statement:
   $\forall n \in \mathbb{N}$ with $n > 1, \forall k \in \{n, n+1, \ldots, 2n\}, \forall (a_1, a_2, \ldots, a_n) \in A(n,k), \exists i \in \{1, 2, ..., n-1\}, a_i = a_{i+1}$

Work: Negate ·n>1, ·k·{n..2n}, ·(a···a·)·A(n,k), ·i, a·=a···
Push ¬ inward: flip ALL quantifiers, negate predicate: ·n>1, ·k·{n..2n}, ·(a··a·)·A(n,k), ·i, a··a··· · (C) ·

   (A) $\exists n \in \mathbb{N}$ with $n > 1, \exists k \in \{n, n+1, \ldots, 2n\}, \exists (a_1, a_2, \ldots, a_n) \in A(n,k), \exists i \in \{1, 2, ..., n-1\}, a_i \neq a_{i+1}$

   (B) $\exists n \in \mathbb{N}$ with $n > 1, \forall k \in \{n, n+1, \ldots, 2n\}, \exists (a_1, a_2, \ldots, a_n) \in A(n,k), \forall i \in \{1, 2, ..., n-1\}, a_i = a_{i+1}$

   (C) $\exists n \in \mathbb{N}$ with $n > 1, \exists k \in \{n, n+1, \ldots, 2n\}, \exists (a_1, a_2, \ldots, a_n) \in A(n,k), \forall i \in \{1, 2, ..., n-1\}, a_i \neq a_{i+1}$

   (D) $\exists n \in \mathbb{N}$ with $n > 1, \exists k \in \{n, n+1, \ldots, 2n\}, \forall (a_1, a_2, \ldots, a_n) \in A(n,k), \forall i \in \{1, 2, ..., n-1\}, a_i \neq a_{i+1}$

   (E) None of the above

· ANSWER

Work: 'For all y' = ∀y, 'exists x' = ∃x, 'exists z' = ∃z
predicate: y = zx² · ∀y, ·∃x, ·∃z, y=zx² · (D) ·

3. For this English statement, find the mathematical statement that corresponds to it: For all real numbers y, there exists another real number x, for which there exists an integer z, such that y is the product of z by x squared.

   (A) $\forall y \in \mathbb{R}, \exists\, x \in \mathbb{R}, \exists z \in \mathbb{Z}, x = zy^2$

   (B) $\forall y \in \mathbb{R}, \exists\, x \in \mathbb{R}, \exists z \in \mathbb{Z}, y \neq zx^2$

   (C) $\forall y \in \mathbb{R}, \forall\, x \in \mathbb{R}, \exists z \in \mathbb{Z}, y = zx^2$

   (D) $\forall y \in \mathbb{R}, \exists\, x \in \mathbb{R}, \exists z \in \mathbb{Z}, y = zx^2$

   (E) None of the above

Part B: Sets & Logic

Please check all statements below and check (A) for True and (B) for False.
U is the universe that contains all possible sets, and $\bar{X}$ is the complement of the set X in the universe U.**[16 points]**

(B) FALSE A\A· = A·(A·)· = A·A = A · · in general

(A) TRUE · is associative ·

4. $\forall A \subset U,\ A \setminus A^C = \emptyset$

(B) FALSE |P| = 2³=8 ; 2 (set has 3 distinct elements)

5. The biconditional operator $\Leftrightarrow$ is associative.

(A) TRUE Standard identity ·

6. $|\mathcal{P}(\{1, \{1, 2\}, \{1\}\})| = 2$

(A) TRUE z²+2z+4=(z+1)²+3; z·0 · z+1·1, matches n²+3 for n···

7. $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$

(A) TRUE P·Q · ¬P·Q ·

8. $\{n^2 + 3 : n \in \mathbb{N}\} = \{z^2 + 2z + 4 : z \in \mathbb{Z} \text{ and } z \geq 0\}$

(A) TRUE P=T · P·Q=T for any Q ·

9. $(P \Rightarrow Q)$ is equivalent to $(\neg P \vee Q)$

(B) FALSE ¬(P·¬Q) = ¬P·¬¬Q = ¬P·Q (not ¬P·Q)

10. If P is true, then $P \vee Q$ is true

11. $\neg(P \wedge \neg Q)$ is equivalent to $(\neg P \wedge Q)$

(B) FALSE 0·{{0},{1},{·}}; elements are SETS {0},{1},{·} · none equals the number 0

(A) TRUE

12. $\emptyset \in \{\{0\}, \{1\}, \{\emptyset\}\}$

(B) FALSE |P(A·B)|=2^|A·B|=2^(|A|+|B|-|A·B|)=2^|A|·2^|B|/2^|A·B| ·

13. $|\mathcal{P}(A \cup B)| = \frac{2^{|A|}2^{|B|}}{2^{|A \cap B|}}$

(A) TRUE A·B · A\B=·, not A

14. $(A \subseteq B) \Rightarrow (A \setminus B = A)$

(A) TRUE A·B · A·B=A ·

15. $(A \subseteq B) \Rightarrow (A \cap B = A)$

(A) TRUE Sets are unordered ·

16. If $A = \{1, 2, 3\}$ and $B = \{2, 3, 1\}$, then $A = B$

(B) FALSE B=C·D · A×B=(A×C)·(A×D); exam uses · · FALSE

17. If $B = C \cap D, A \times B = (A \times C) \cup (A \times D)$

(B) FALSE |{·}|=1 not 0 (contains one element: the empty set)

18. $|\{\emptyset\}| = 0$

(B) FALSE B={2,{2}}: needs {2}·A but A={2,{2,{2}}} has no {2} element

19. Let $A = \{2, \{2, \{2\}\}\}$ and $B = \{2, \{2\}\}$. Is $B \subseteq A$?

Part C: Counting

The Cruz-Hack Club organized a hackathon. Four teams participated (Cruzers, K-Combo, CodeCode, Sluggies), for a total of 11 student participants (that is a set of hackers $H = \{h_1, h_2, \ldots, h_{11}\}$ and five judges (that is, a set $J = \{j_1, j_2, \ldots, j_5\}$) were invited to rate the projects. When voting, votes are anonymous.

20. Assuming all the students and the judges enter the Hackathon room one by one, how many ways are there for them to enter in such a way that no two judges follow each other (that is, in such a way that there is at least one student in between the judges). **[2 points]**

NAME: Student ID:

11! × C(12,5) × 5!

Method: Place 11 students first · 11! ways.
11 students create 12 gaps (_, S, _, S, ..., S, _).
Choose 5 gaps for 5 judges: C(12,5) ways.
Arrange 5 judges in chosen gaps: 5! ways.
Total = 11! × C(12,5) × 5! [guarantees no two judges adjacent]

ANSWER:_____

21. In the room, there are 16 tables for the students to sit at. Each table sits 2 people. How many students would we need to ensure that 2 students sit at the same table? **[2 points]**

ANSWER:_____

17

16 tables, each sits 2. Worst case: 1 student per table = 16 students, no pair.
One more (17th) forces ·2 at some table. Pigeonhole: n+1 pigeons, n holes.
Answer: 16 + 1 = 17

22. For some awards, all participants and judges (for a total of 16 votes) get to vote. How many ways are there to split the vote among the four projects, considering it is possible a project receives no votes? **[2 points]**

ANSWER:_____

C(19, 3) = 969

Stars & bars, 0 allowed per team. n=16 votes, k=4 teams.
Formula: C(n+k-1, k-1) = C(16+3, 3) = C(19,3)
C(19,3) = 19!/(3!·16!) = (19·18·17)/6 = 969

23. One of the project of the Hackathon answers this question: given k distinct natural numbers $a_1, a_2, \ldots, a_k$ between 1 and 32, what is the minimum value of k such that there are two consecutive numbers in the set $\{a_1, a_2, \ldots, a_k\}$? **[2 points]**

17

Max set with no two consecutive in {1..32}: take all odds {1,3,5,...,31} = 16 numbers.
Any 17th number from {1..32} creates a consecutive pair (every even is adj to odd).
Min k = 16 + 1 = 17

ANSWER:_____

24. Assuming 16 people will vote for any of the 4 teams again, how many votes do we need to count to guarantee that at least one team has received three votes? [**2 points**]

    ANSWER:_____

9

Worst case: distribute votes so no team reaches 3.
Give each of 4 teams exactly 2 votes = 8 votes total · no team has 3.
One more vote (9th) forces some team to ·3. Answer: 8+1 = 9

25. Assuming the 16 voters can vote for any team, what can you say about the minimum number of votes to count such that all team receive at least one vote? [**2 points**]

    ANSWER:_____

Cannot be guaranteed.

Voters choose freely. All 16 votes could go to one team (16, 0, 0, 0).
No matter how many total votes, the distribution could skip any team entirely.
It is IMPOSSIBLE to guarantee each team gets ·1 vote without controlling distribution.

26. The organizers do not want a team to have no vote. They collude with the judges in order to achieve this outcome. How many ways are there to split the 16 votes such that each of the 4 teams has at least one vote? [**2 points**]

NAME: Student ID:

C(15, 3) = 455

Stars & bars, each team gets ·1 vote. n=16, k=4, each ·1.
Give 1 vote to each team first (4 used). Distribute 12 remaining freely? No ·
use direct formula: C(n-1, k-1) = C(16-1, 4-1) = C(15,3).
C(15,3) = 15!/(3!·12!) = (15·14·13)/6 = 455

ANSWER:

27. For some other award, the organizers do not want all the participants to vote, but only the judges and 5 students. How many ways are there to select the 5 voting students among all 11 students? **[2 points]**

ANSWER:_____

C(11, 5) = 462

Select 5 voting students from 11. Unordered, no repetition · C(n,r).
C(11,5) = 11!/(5!·6!) = (11·10·9·8·7)/(5·4·3·2·1) = 462

28. For one of the awards, the 16 votes are tallied and read in order. How many possible lists of votes can be read in such manner, considering each vote is one of the four project names, and that the same project can be voted for repeatedly.

ANSWER:_____

4^16

16 votes read in order; each vote independently chooses 1 of 4 teams.
Ordered sequence with repetition allowed · n· = 4^16.
4^16 = 4,294,967,296

29. The organizers will select three participants to represent the Cruz-Hack Club at the state level. They will select 3 students from both the winning team and the runner-up team. Both these teams have four members each. The organizers ensure that at least one of the selected students is from the winning team, and the other two selected students can come from either team. How many ways are there to select these 3 students? **[2 points]**

52

Winning team: 4 members. Runner-up team: 4 members. Total pool: 8 students.
Select 3 with ·1 from winning. Use complement:
Total ways to choose 3 from 8: C(8,3) = 56
All 3 from runner-up only (0 winners): C(4,3) = 4
At least 1 from winning: 56 - 4 = 52

ANSWER:

Part D: Truth Table

Write the truth table to answer the following question: Is $(P \wedge (Q \Leftrightarrow R)) \Rightarrow (\neg P \Rightarrow (Q \vee R))$ a tautology, that is a proposition that is always true. The truth table should have columns for (1): $(Q \Leftrightarrow R)$, (2): $(Q \vee R)$, (3): $(P \wedge (Q \Leftrightarrow R))$, (4): $(\neg P \Rightarrow (Q \vee R))$ and (5): $(P \wedge (Q \Leftrightarrow R)) \Rightarrow (\neg P \Rightarrow (Q \vee R))$ (2 points per column). Based on the truth table, state your answer to the question. **[10 points]**

P	Q	R	Q·R	Q·R	P·(Q·R)	¬P·(Q·R)	(P·(Q·R))· (¬P·(Q·R))
T	T	T	T	T	T	T	T
T	T	F	F	T	F	T	T
T	F	T	F	T	F	T	T
T	F	F	T	F	T	T	T
F	T	T	T	T	F	T	T
F	T	F	F	T	F	T	T
F	F	T	F	T	F	T	T
F	F	F	T	F	F	F	T

Conclusion: Column (5) is TRUE in all 8 rows.
· The statement (P·(Q·R)) · (¬P·(Q·R)) is a TAUTOLOGY.